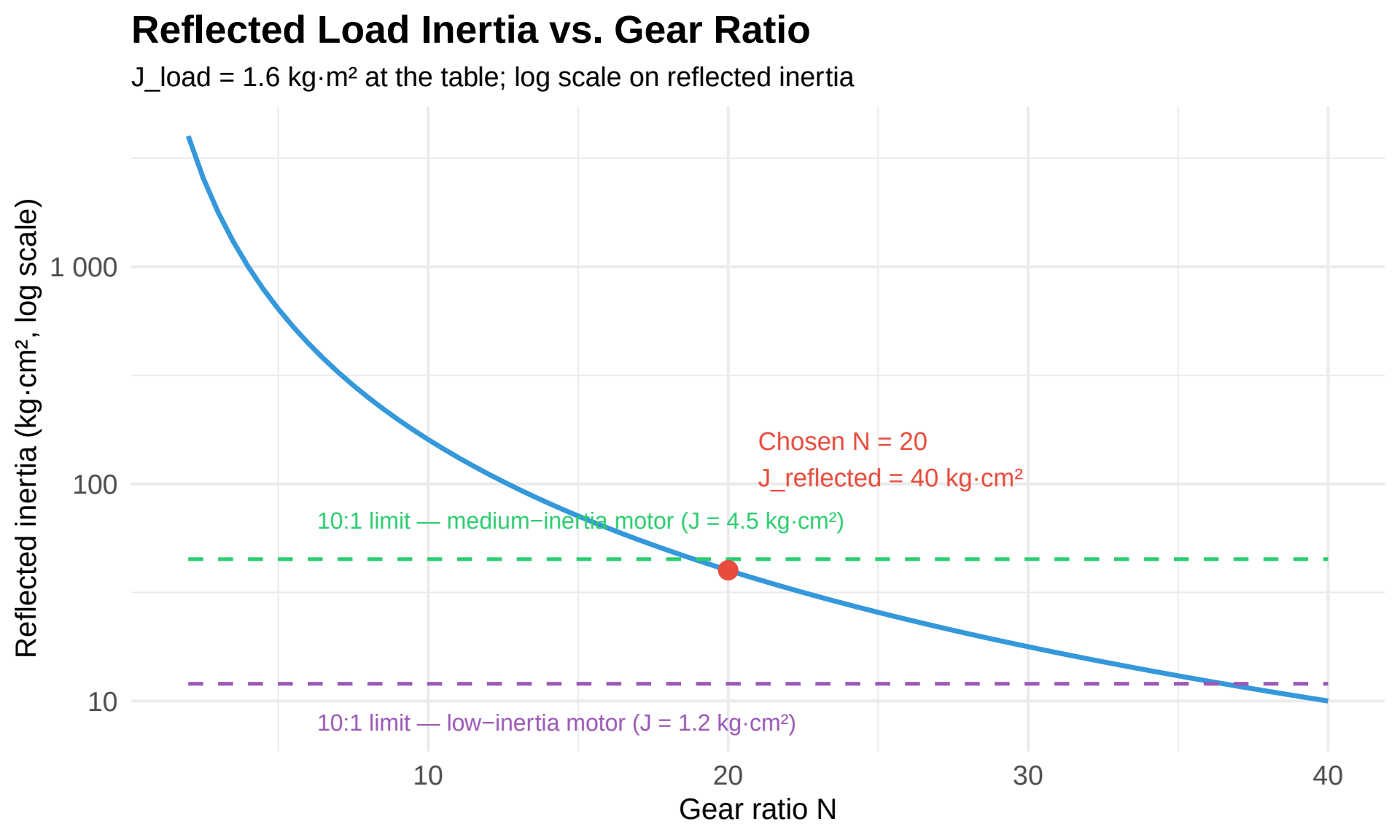
Reflected Load Inertia vs. Gear Ratio
J_load = 1.6 kg·m² at the table; log scale on reflected inertia
Reflected inertia (kg·cm², log scale)
1 000
100
10
Chosen N = 20
J_reflected = 40 kg·cm²
10:1 limit — medium−inertia motor (J = 4.5 kg·cm²)
10:1 limit — low−inertia motor (J = 1.2 kg·cm²)
10
20
30
40
Gear ratio N